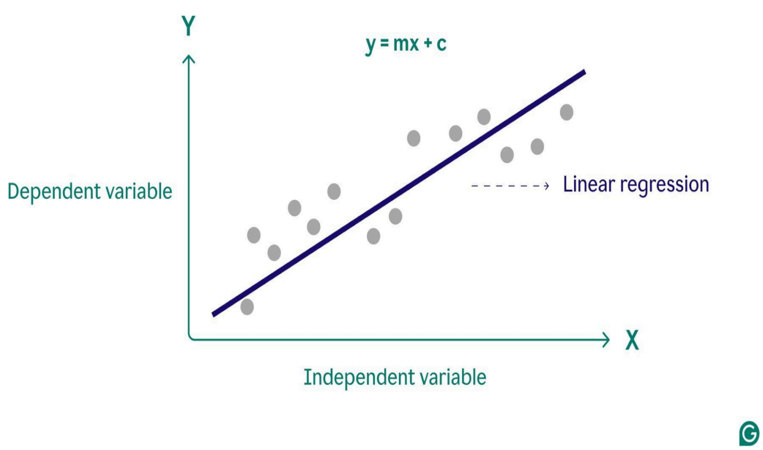
Overview of Classification Algorithms in Machine Learning

1.Linear Regression:

Description:

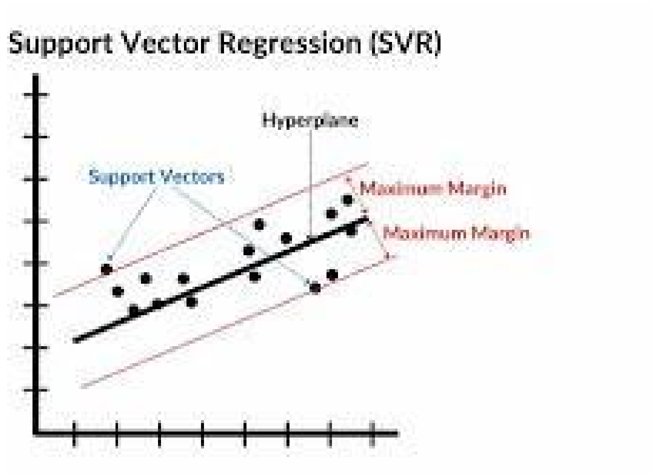
Linear regression is a supervised machine learning algorithm and statistical method that models the linear relationship between a dependent variable (y) and one or more independent variables (x) by fitting a straight "best-fit" line, calculated using the least-squares method.



2. Support Vector Regression (SVR):

Description:

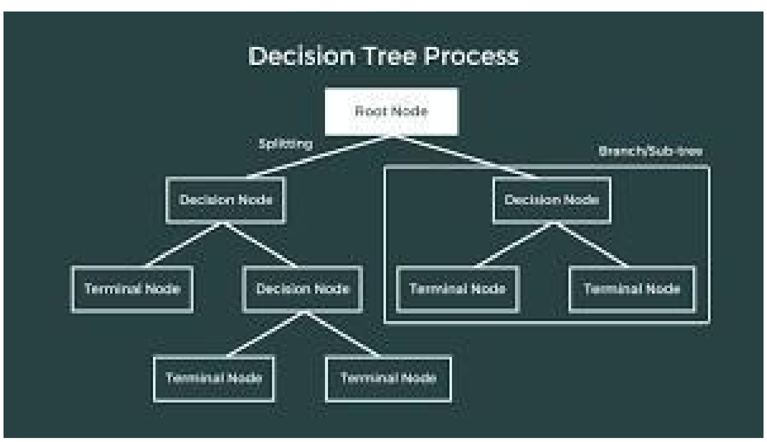
Support Vector Regression (SVR) is a supervised learning algorithm that predicts continuous values by finding a "tube" (hyperplane) that best fits the data within a specified tolerance (\epsilon \)-insensitive loss function). Unlike traditional regression, SVR focuses on minimizing error while maximizing the margin of fit.



3. Decision Tree Regressor:

Description:

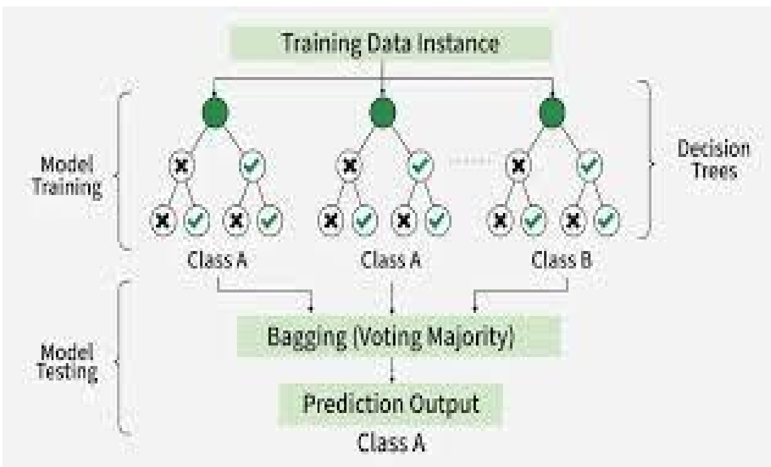
A Decision Tree Regressor splits data into branches based on feature values. Each split reduces error and leads to a final prediction at the leaf node. It is easy to understand but may overfit if the tree becomes too deep.



4. Random Forest Regressor:

Description:

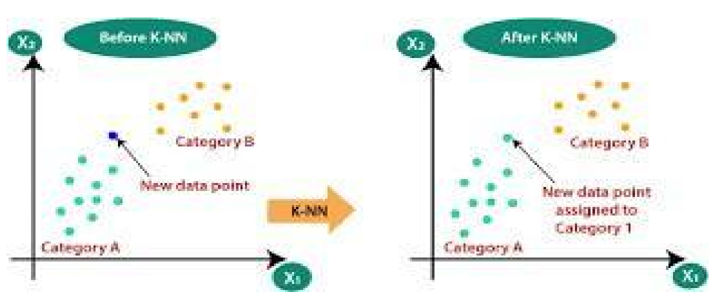
Random Forest is an ensemble method that combines multiple decision trees. Each tree gives a prediction, and the final output is the average of all. It reduces overfitting and improves accuracy, making it one of the best models for this dataset.



5. K-Nearest Neighbors (KNN) Regressor:

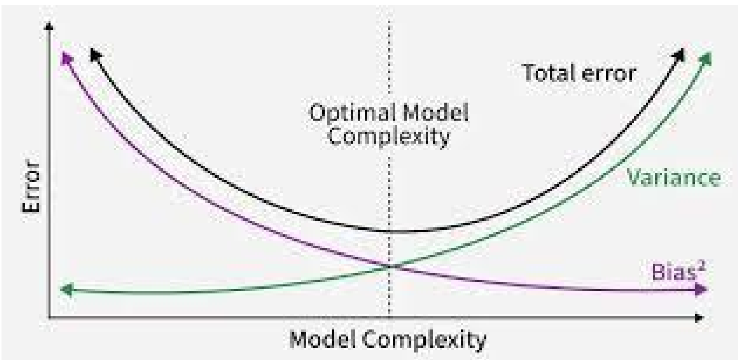
Description:

KNN predicts house price based on the average of nearest similar houses. It uses distance (like Euclidean distance) to find similar data points. It is simple but slower for large datasets.



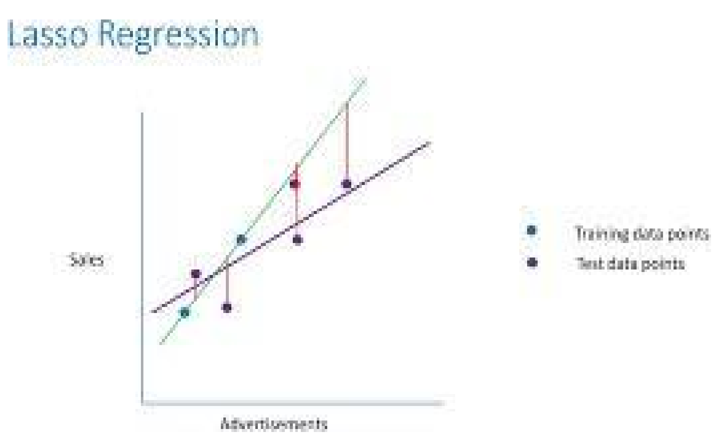
6. Ridge Regression:
Description:

Ridge Regression is an improved version of Linear Regression. It adds a penalty (regularization) to reduce overfitting. It works well when features are correlated and improves model stability.



7. Lasso regression:
Description:

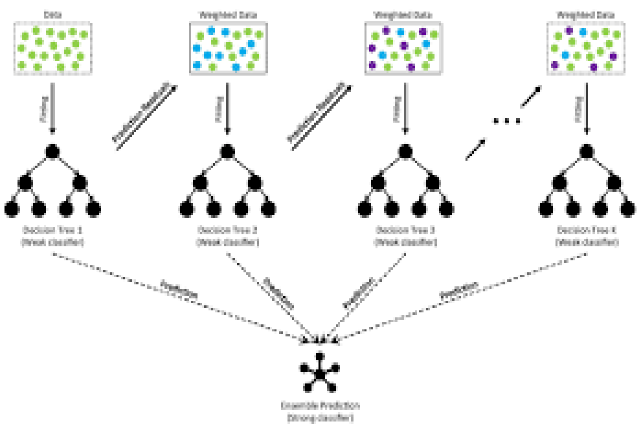
Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a linear regression technique that uses L1 regularization to enhance model prediction accuracy and interpretability.



8 .Gradient Boosting Regression

Description:

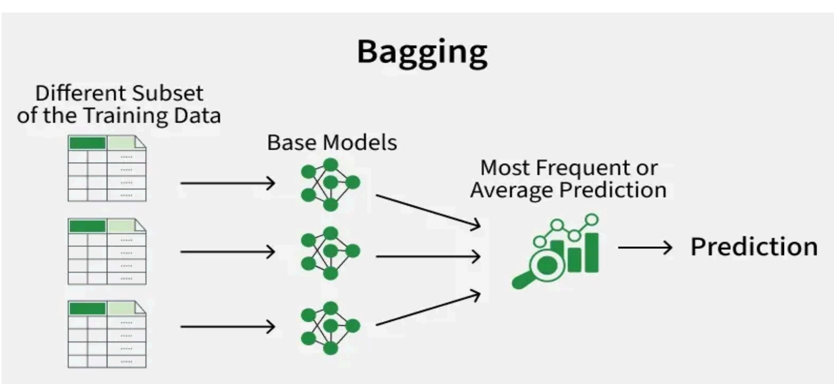
Gradient Boosting Regression is an ensemble machine learning algorithm that builds a strong predictive model by combining multiple weak learners, typically decision trees. It works by training models sequentially, where each new model focuses on correcting the errors made by the previous models. The algorithm minimizes the loss function using a technique called gradient descent.



9.Bagging Regression (Bootstrap Aggregating)

Description:

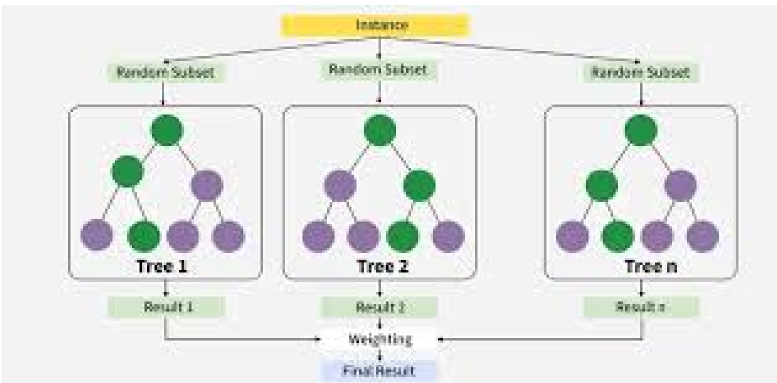
Bagging Regression is an ensemble learning technique that improves model performance by training multiple models on different random subsets of the dataset (created using bootstrap sampling) and then combining their predictions by averaging. It is mainly used to reduce variance and avoid overfitting.



10.XGBoost Regression (Extreme Gradient Boosting)

Description:

XGBoost (Extreme Gradient Boosting) is an advanced and efficient implementation of gradient boosting that uses optimized algorithms, parallel processing, and regularization techniques to improve performance and prevent overfitting. It is widely used in machine learning competitions and real-world applications due to its high speed and accuracy.



Final Conclusion:

In this project, different regression algorithms were applied to predict house prices using features like bedrooms, bathrooms, square footage, and more. Each model performs differently based on its learning technique. Among all models, ensemble methods like Random Forest and Lasso regression generally give better accuracy because they combine multiple learners and reduce errors.

What I have Learned:

- I learned how machine learning can be used for real-world prediction problems like house prices.
- I understood different regression algorithms and how they work.
- I learned the importance of feature scaling and train-test splitting.
- I understood how models are compared using performance metrics.
- I learned that ensemble methods give better performance than simple models.
- Overall, I gained practical knowledge of building and evaluating a complete machine learning pipeline.